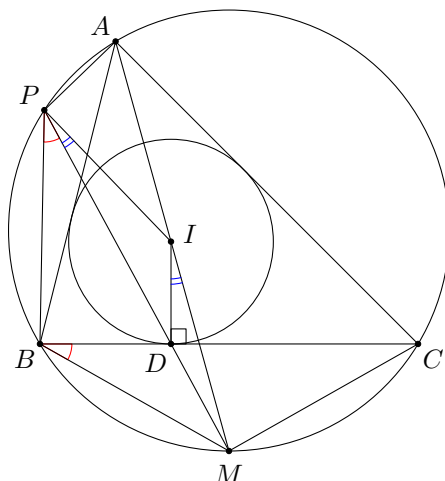


Problem 5 (EGMO 2014 Problem 2). Let ABC be a triangle with $AB \neq AC$ and circumcircle Ω . Its incircle has centre I and touches the side BC at point D . Line AI intersects Ω at M again, and DM intersects Ω again at P .

Prove that $\angle API = 90^\circ$.



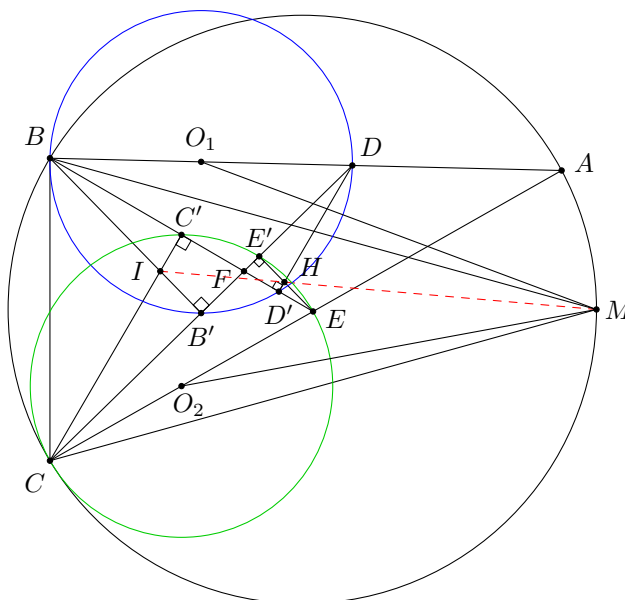
Solution. WLOG assume $AB < AC$. Note that M is the midpoint of arc BC of Ω . So $\angle MBC = \angle BPM$. This implies $\triangle MBD \sim \triangle MPB$, and hence $MB^2 = MD \times MP$. It follows that $MI^2 = MB^2 = MD \times MP$, and thus $\angle DIM = \angle MPI$. Then we have

$$\angle IPA = \angle MPA - \angle MPI = 180^\circ - \angle ACM - \angle DIM = 180^\circ - \left(C + \frac{A}{2}\right) - \left(90^\circ - C - \frac{A}{2}\right) = 90^\circ.$$

□

Problem 6. Let ABC be a triangle with points D and E on segments AB and AC , respectively, such that $DB = BC = CE$. Let the lines CD and BE meet at F .

Prove that the incentre I of triangle ABC , the orthocentre H of triangle DEF and the midpoint M of the arc BAC of the circumcircle of triangle ABC are collinear.



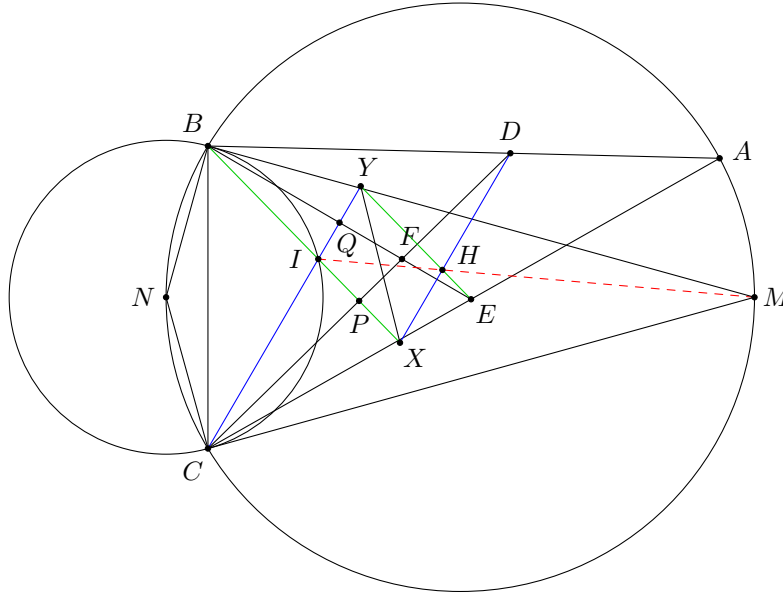
Solution 1. Let BI meet CD at B' , CI meet BE at C' , DH meet EF at D' , and EH meet DF at E' . Note that $BB' \perp CD$, $CC' \perp BE$, $DD' \perp EF$ and $EE' \perp DF$. Let $\Gamma_1 = (BB'D'D)$ and $\Gamma_2 = (CC'E'E)$.

Since B, C, B', C' are concyclic, we have $IB \times IB' = IC \times IC'$. So I lies on the radical axis of Γ_1 and Γ_2 .

Since D, E, D', E' are concyclic, we have $HD \times HD' = HE \times HE'$. So H lies on the radical axis of Γ_1 and Γ_2 .

Let O_1 and O_2 be the centre of Γ_1 and Γ_2 respectively. So they are the midpoints of BD and CE . Since $MB = MC$, $BO_1 = CO_2$ and $\angle MBO_1 = \angle MCO_2$, we have $\triangle MBO_1 \cong \triangle MCO_2$. This gives $MO_1^2 - BO_1^2 = MO_2^2 - CO_2^2$, i.e. the power of M with respect to Γ_1 and Γ_2 are equal. Hence, M lies on the radical axis of Γ_1 and Γ_2 .

It follows that I, H, M are collinear. □



Solution 2. Let N be the other midpoint of arc BC . Let DH meet BI at X , and let EH meet CI at Y .

Note that N is the centre of (BCI) . As $\angle NBM = \angle NCM = 90^\circ$, MB and MC are tangents to (BCI) . So MI is the I -symmedian of $\triangle BIC$.

Since $BC = BD$, we have $BI \perp CD$. This yields $BI \parallel YE$, and similarly $CI \parallel XD$. So $IXHY$ is a parallelogram. The diagonal IH passes through the midpoint of XY , i.e. IH is the median of $\triangle IXY$.

Let BI meet CD at P , CI meet BE at Q . As $BP \perp CF$ and $CQ \perp BF$, we have $\triangle IBC \sim \triangle IQP$. Also, note that $CXDI$ is a rhombus. So P is the midpoint of IX . Similarly, Q is the midpoint of IY . It follows that $\triangle IBC \sim \triangle IQP \sim \triangle IYX$. Then the median IH of $\triangle IXY$ is just the symmedian of $\triangle IBC$.

Therefore, I, H, M are collinear. □

Problem 7. Two points P and Q lying on side BC of triangle ABC and their distance from the midpoint of BC are equal. The perpendiculars from P and Q to BC intersect AC and AB at E and F , respectively. Let M be point of intersection PF and EQ and let H_1 and H_2 be the orthocentres of triangles BFP and CEQ , respectively.

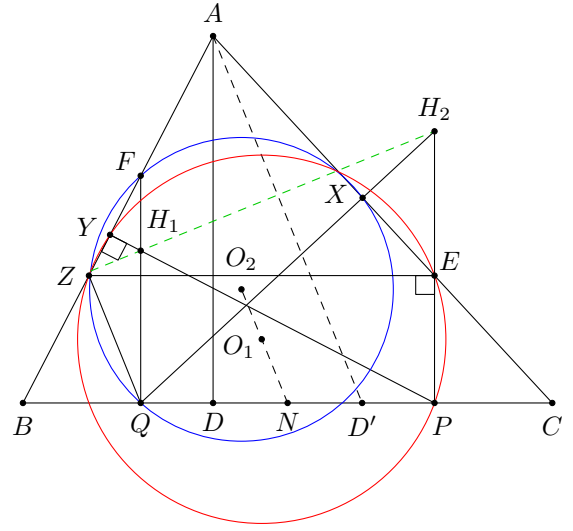
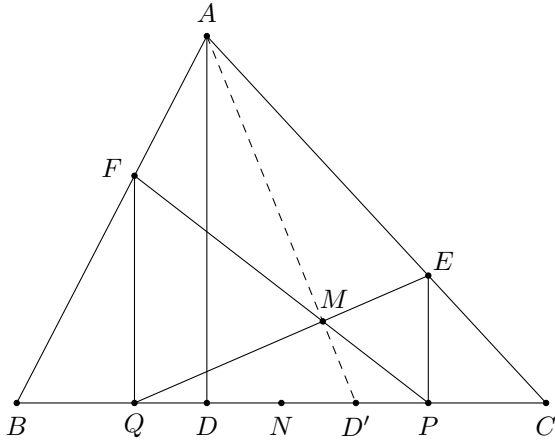
Prove that $AM \perp H_1H_2$.

Solution. Let N be the midpoint of BC , let D be the foot of altitude from A , and let D' be the symmetric point of D about N . We shall prove that M always lies on AD' , while H_1H_2 is always perpendicular to AD' , which clearly implies the result.

Firstly, note that $FQ \parallel AD \parallel EP$, so we have $\triangle BQF \sim \triangle BDA$, $\triangle CPE \sim \triangle CDA$ and $\triangle MFQ \sim \triangle MPE$. As $BD' = CD$, $D'P = DQ$ and $PC = QB$, we have

$$\frac{PM}{MF} \times \frac{FA}{AB} \times \frac{BD'}{D'P} = \frac{PE}{QF} \times \frac{QD}{DB} \times \frac{CD}{DQ} = \frac{PE}{DA} \times \frac{DA}{QF} \times \frac{CD}{DB} = \frac{PC}{DC} \times \frac{DB}{QB} \times \frac{CD}{DB} = 1.$$

By Menelaus' theorem, A, M, D' are collinear.



Secondly, let X and Y be the projection of Q and P onto AC and AB respectively. Note that F, Y, Q, P are concyclic, and so $H_1F \times H_1Q = H_1Y \times H_1P$. Hence H_1 lies on the radical axis of (PEY) and (QFX) . Similarly, since E, X, Q, P are concyclic, H_2 also lies on this radical axis. Let O_1 and O_2 be the centres of (PEY) and (QFX) respectively. Since O_1O_2 is perpendicular to the radical axis H_1H_2 , it remains to prove $O_1O_2 \parallel D'A$. In fact, we will show that both O_1 and O_2 lie on the line through N parallel to $D'A$.

Let Z be the point on AB such that $ZE \parallel BC$. As $\angle ZEP = 90^\circ = \angle ZYP$, the point Z lies on (PEY) , and PZ is a diameter of this circle. Thus, O_1 is the midpoint of PZ . Since

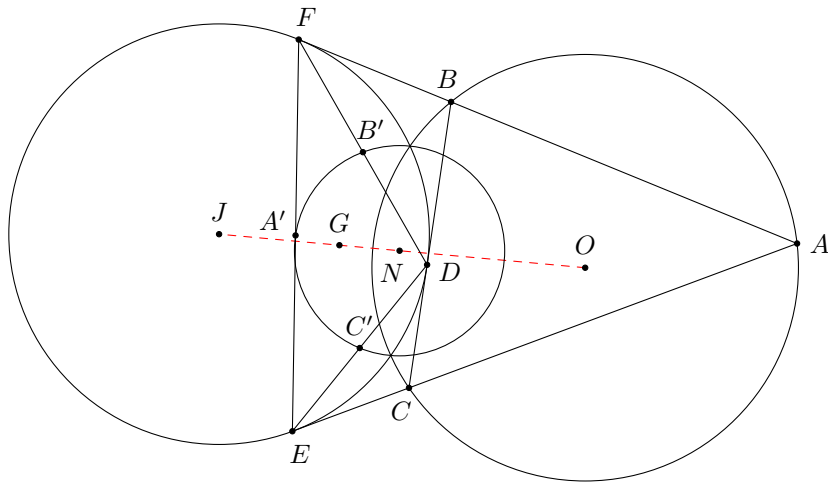
$$\frac{BZ}{ZA} = \frac{CE}{EA} = \frac{CP}{PD} = \frac{BQ}{QD'},$$

we have $ZQ \parallel AD'$. By the midpoint theorem, we have $O_1N \parallel ZQ$, and so $O_1N \parallel AD'$. The proof is complete. $\square$

Problem 8. Given two fixed circles ω_1, ω_2 , let $\mathcal{S}$ denote all $\triangle ABC$ satisfying the following conditions:

- ω_1 is the circumcircle of $\triangle ABC$,
- ω_2 is the A -excircle of $\triangle ABC$ which touches BC, CA, AB at D, E, F , respectively, and
- $\mathcal{S}$ is non-empty.

Prove that the centroid of $\triangle DEF$ is a fixed point.



Solution. Let O and J be the centre of ω_1 and ω_2 respectively. Consider the inversion with respect to ω_2 . The images A', B' and C' of A, B and C are the midpoints of EF, FD and DE respectively. So the image of ω_1 is $(A'B'C')$, which is the nine-point circle of $\triangle DEF$. As this image is fixed, the nine-point centre N of $\triangle DEF$ is fixed. Note that J is the circumcentre of $\triangle DEF$. So the centroid G of $\triangle DEF$ lies on the segment JN in the ratio $JG : GN = 2 : 1$. Thus, G is a fixed point. $\square$